

BlindSpot Barnacles

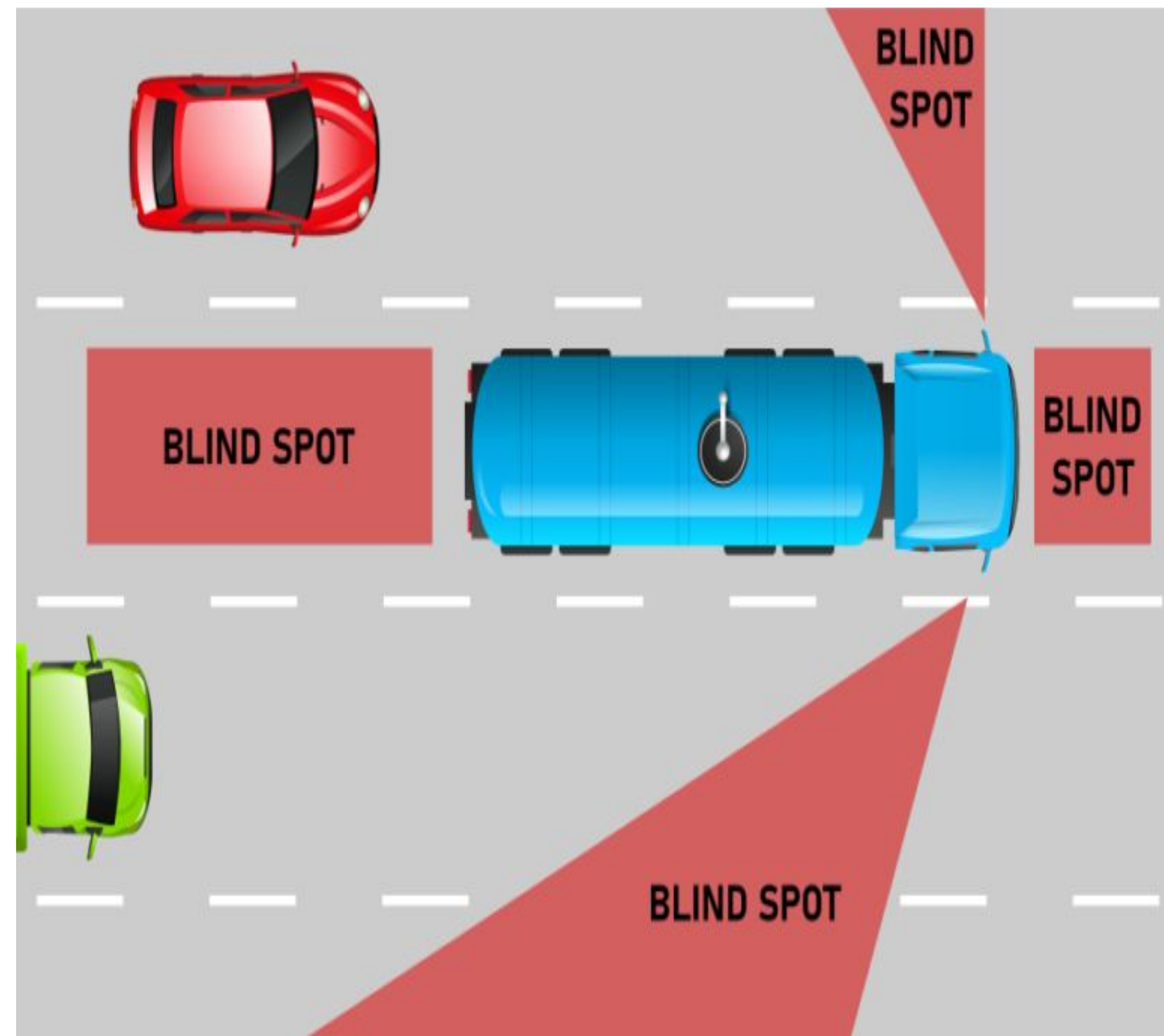
Blind Spot Detection System for Semi-Trucks

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Ever Been Cut-Off by a Semi?

Roughly $\frac{1}{3}$ of crashes between a car and a large truck happen inside of blind spots.

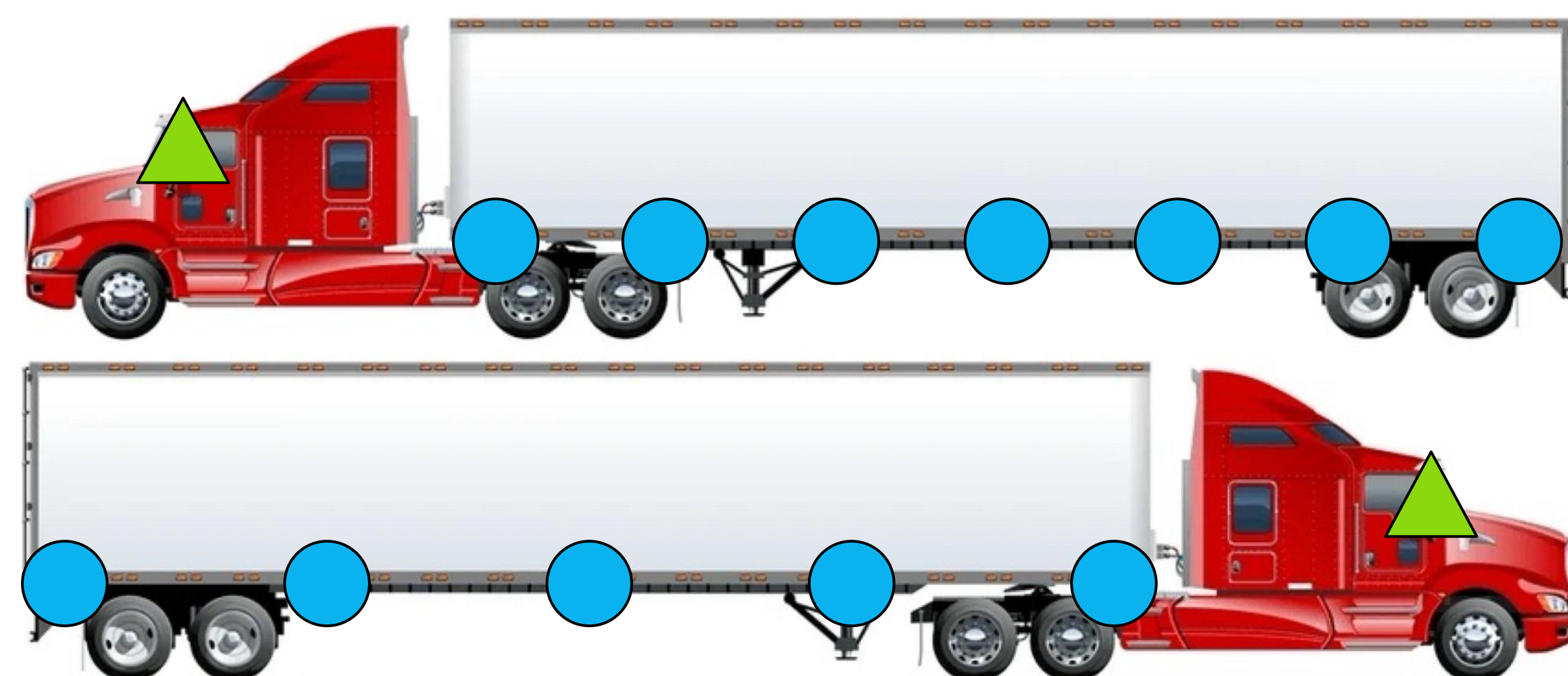
(Federal Motor Carrier Safety Administration)



Our Solution

BlindSpot Barnacles provide a detachable blind spot detection system, allowing them to be used on cars as well as semi-trucks.

▲ Indicator ● Barnacle



How it Works

App

Barnacle Connect is a free app for users to setup BlindSpot Barnacles.



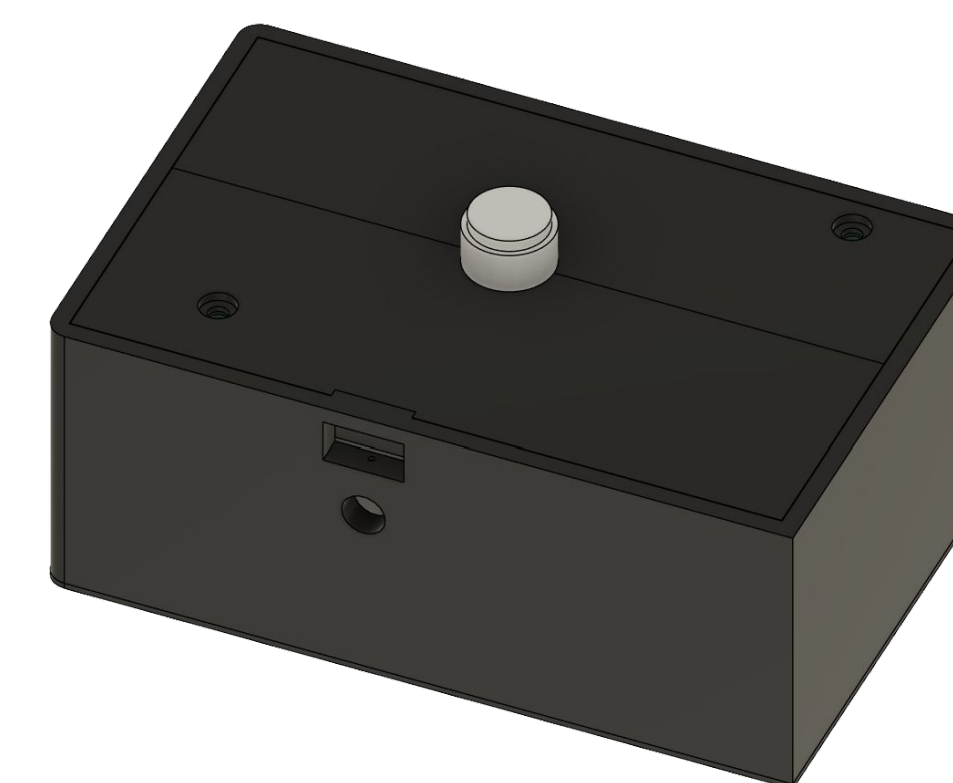
Users can:

- Configure orientation
- View connection status

Barnacle

Components:

- Magnet mounts to exterior
- Battery-powered
- USB-C charging
- ESP32 dev board



Functionality:

- Ultrasonic sensor detects object
- Bluetooth sends signal to indicators

Indicator

Components:

- Sticker mounts to interior trim
- Battery-powered
- USB-C charging
- ESP32 dev board

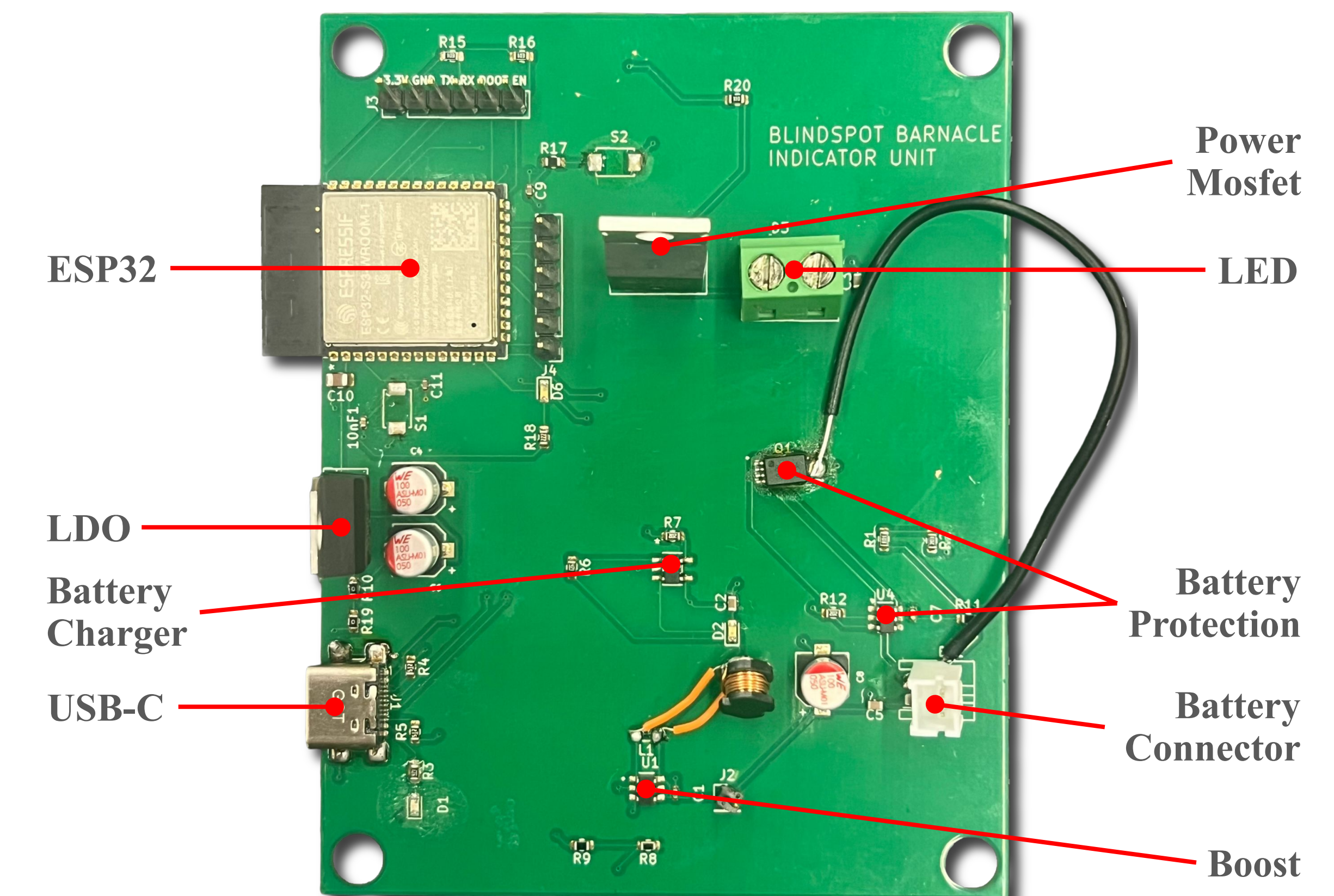


Functionality:

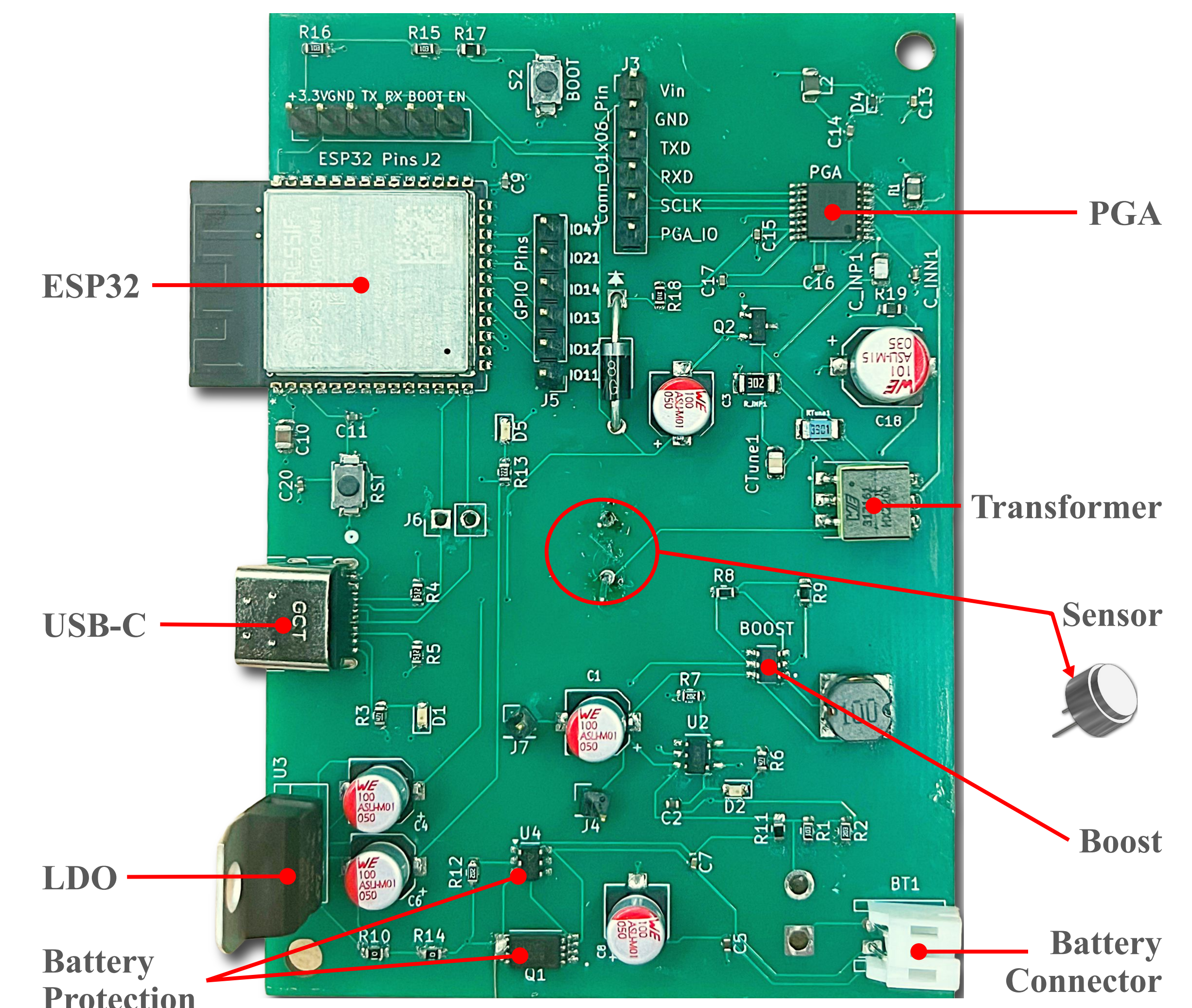
- Bluetooth receives signal
- LED lights up

PCB Overview

Indicator



Barnacle



Acknowledgements

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